Model-Heterogeneous Semi-Supervised Federated Learning for Medical Image Segmentation

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***Abstract*— Medical image segmentation is crucial in clinical diagnosis, helping physicians identify and analyze medical conditions. However, this task is often accompanied by challenges like sensitive data, privacy concerns, and expensive annotations. Current research focuses on personalized collaborative training of medical segmentation systems, ignoring that obtaining segmentation annotations is time-consuming and laborious. Achieving a perfect balance between annotation cost and segmentation performance while ensuring local model personalization has become a valuable direction. Therefore, this study introduces a novel Model-Heterogeneous Semi-Supervised Federated (HSSF) Learning framework. It proposes Regularity Condensation and Regularity Fusion to transfer autonomously selective knowledge to ensure the personalization between sites. In addition, to efficiently utilize unlabeled data and reduce the annotation burden, it proposes a Self-Assessment (SA) module and a Reliable Pseudo-Label Generation (RPG) module. The SA module generates self-assessment confidence in real-time based on model performance, and the RPG module generates reliable pseudo-label based on SA confidence. We evaluate our model separately on the Skin Lesion and Polyp Lesion datasets. The results show that our model performs better than other methods characterized by heterogeneity. Moreover, it exhibits highly commendable performance even in homogeneous designs, most notably in region-based metrics. The full range of resources can be readily accessed through the designated repository located at *HSSF(github.com)* on the platform of GitHub.**

***Index Terms*— Federated learning, knowledge distilling, medical image segmentation, semi-supervised learning.**

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I. Introduction

DEEP learning has made remarkable strides in lesion segmentation [1], [2], organ segmentation [3], [4], instrument segmentation [5], [6], etc. Despite such progress, prior research predominantly emphasizes the enhancement of segmentation metrics, neglecting certain practical difficulties that incorporating data solely from a single source could potentially result in overfitting issues and undermine clinical applicability. While implementing a centralized training approach to collect multi-institution data is impracticable due to data silos and concerns about data privacy.

Federated Learning [7] is proposed to address data privacy protection by training models at local sites and aggregating the local parameters rather than local data at a cloud server, as exemplified in Fig. 1. However, when applied to medical tasks, vanilla Federated Learning methods still suffer from considerable heterogeneity across unique local sites caused by variations in imaging protocols and operation customs. Two types of heterogeneity are commonly encountered: data heterogeneity and model heterogeneity. Data heterogeneity refers to variations in the local datasets of each site, including differences in data distribution, type, volume, and quality. Balancing these diverse contributions during global model aggregation is essential to prevent skewed influences on the overall model performance. Model heterogeneity involves differences in the local models used by sites, arising from variations in their datasets or choices of model architecture, hyperparameters, or training methods. Managing these model disparities is a key challenge, as the global model must effectively integrate updates from diverse sites with varying capabilities and convergence rates. Based on knowledge distillation mechanisms [8], Model-Heterogeneous Federated Learning (HFL) methods are proposed, where each local model obtains a distinct network architecture to fit the local distribution [9], [10], [11], [12], [13]. Nevertheless, existing HFL methods rarely notice the heterogeneity in annotations that were acquiring annotations for segmentation is arduous and time-intensive undertaking. Due to the supervised setup in most existing paradigms, local clients without expert annotations are unable to join and effectively contribute to the Federated Learning process. Models with only a small number of annotations tend to experience overfitting, which in turn prevents global models from learning truly useful and applicable knowledge in practice.

Semi-Supervised Learning is seen to be a useful method for reducing training annotations since it relies on a significant amount of unlabeled data to obtain excellent results for models with only a small fraction of labeled data in the target dataset. The present investigation on Semi-Supervised Learning may be broadly categorized into the self-training-based approaches [14], [15] and the consistency-based regularization approaches [16], [17]. The former entails an initial step where the model ascribes pseudo-labels to the unlabelled data, which it subsequently utilizes to supervise the learning process. The latter employs constraints to enforce consistency among the outputs generated by numerous perturbation inputs, thereby ensuring that the decision boundary resides within a low-density region. However, these studies often employ centralized training settings and do not consider how to leverage collaborative training across multiple sites. Due to disparities in the local environments and model structures at different locations, when simply integrated into the HFL framework, the effectiveness of these methods may even fall short of local training. Furthermore, suboptimal performance by certain models during the knowledge aggregation process can potentially lead to a decline in the overall framework performance.

In light of the above concerns, we propose a Federated Semi-Supervised Learning approach with model heterogeneity. Unlike vanilla Federated Learning methods, HSSF grants substantial autonomy to individual sites, allowing them to independently design model structures. During training, instead of rigidly learning from fixed parameters, HSSF enables sites to selectively learn according to their specific requirements.

Additionally, HSSF goes beyond leveraging knowledge from other sites through Federated Learning; it also eases the annotation burden by incorporating Semi-Supervised Learning principles. Since site data does not require labeling, this approach lowers the barrier to participation in training by other sites. By effectively combining the strengths of Federated Learning and Semi-Supervised Learning, HSSF not only reduces the annotation workload but also maximizes the utilization of local unlabeled data resources. Furthermore, it ensures the security of local data, creating a mutually beneficial scenario.

During local training, the **SA** module of the local model conducts a self-assessment of prediction results alongside model output segmentation predictions. Subsequently, the **RPG** module is employed to filter segmentation predictions based on the **SA** outcomes, enhancing the reliability of pseudo-labels and improving the effectiveness of local training. Exceptional local models can contribute additional knowledge to enhance global training.

To comprehensively demonstrate the advantages of HSSF, we not only compare it with state-of-the-art (SOTA) heterogeneous methods but also conduct comparisons in a homogeneous setting. We select some SOTA Semi-Supervised methods, both existing and our proposed approach, and integrate them with FedAvg [7] for comparison purposes in homogeneous scenarios. This enables us to showcase the superiority of our Semi-Supervised approach.

The principal advancements of this research can be summarized as the following:

- We innovatively divide the knowledge transfer of Heterogeneous Federated Learning into two distinct phases, namely, *Regularity Condensation* and *Regularity Fusion*. In each communication epoch, the server first condenses the Regulations that are closest to the truth among all sites and then broadcasts these Regulations to all sites, which wisely chooses some of the beneficial Regulations for fusion.
- We propose the **SA** module and **RPG** module to correct pseudo-label inaccuracies. The **SA** module takes intermediate activation and prediction graphs as cues to obtain self-assessment confidence. The **RPG**, on the other hand, adaptively corrects pseudo-labels based on **SA** confidence.
- We set up two medical scenarios using one skin lesion segmentation dataset and five polyp segmentation datasets, respectively. Also, we compared the model with various superior methods under the assumptions of Heterogeneous and Homogeneous Federated Learning, respectively. The experimental results show that the model has SOTA performance in all considered cases.

II. Related Works

A. Federated Learning

Federated Learning is a machine learning paradigm that facilitates the collaborative training of models among clients, thereby safeguarding data privacy. In their setup, the server uses average model parameters for the collective consolidation of the local model. Wang et al. [18] proposed the Federated Matching Average algorithm (FedMA), which is an effective method for Federated Learning within the framework of modern neural networks by building a globally shared model sequentially layer by layer. Li et al. [19] applied local batch normalization before feature transfer to alleviate the limitations of heterogeneous data distributions. Wang et al. [20] addressed the Heterogeneous Medical Image Segmentation problems by exploring the inter-site differences.

Although these studies have yielded good results, they required the same structure in each local site and have ignored the individualized needs of customers due to **structural constraints**. The following work instead attempts to tailor a unique model for individual clients. Arivazhagan et al. [21] divided the model into a base layer and a personalization layer. The base layer is trained by parameter averaging, while the personalized layer using different structures is trained based on local data using stochastic gradient descent. Experimentally, this approach proved to be effective in improving the personalization of Federated Learning, while its ability to achieve heterogeneity is considered insufficient. Some studies enable collaborative training of multiple sites through knowledge distillation [8].

Li and Wang [9] proposed using transfer learning and knowledge distillation to enable inter-model communication.

Lin et al. [10] employed unlabeled data generated by the clients' models to facilitate the training of a central classifier. Diao et al. [22] introduced HeteroFL, a method that utilizes knowledge distillation to reduce privacy risks and minimize costs. Furthermore, this technique provides the capability for flexible aggregation of heterogeneous client models. Fang and Ye [12] addressed the heterogeneous model by calculating the optimal weighted aggregate value of the participating customers and dynamically modifying the degree of contribution of individual customers.

These methods effectively enhanced the heterogeneity of the model but they were mostly designed for classification tasks. Federated Learning based on knowledge distillation is expected to effectively address segmentation tasks with individualized settings while facilitating the effective transfer and utilization of client insights and eliminating the need for parameter updates. Different from previous studies, in the process of knowledge transfer, we propose an autonomous selection mechanism, which endows the model with greater autonomy according to reality compared with simple weighted aggregation.

B. Semi-Supervised Learning

The crux of the matter in Semi-Supervised Learning lies in formulating efficiently supervised cues for unlabeled data. The present solutions are primarily bifurcated into two main approaches, namely consistency normalization [16], [17] and self-training-based approach [14], [15]. Consistency regularization is based on the premise that the predictions for unlabeled examples should remain constant under different perturbations. Self-training [23] is a highly regarded and extensively employed technique within the realm of machine learning. By assigning pseudo-labels to unlabeled data, which is subsequently integrated with manually labeled data and further refined by additional rounds of retraining. The assigned pseudo-labels may carry the inherent risk of overconfidence [24], [25]. This means that the model may learn to over-conform to the false confident pseudo-label, negatively impacting model training.

Based on this, Sohn et al. proposed FixMatch [26], which utilizes a combination of the two widely used methods mentioned above. Specifically, FixMatch exploits the model's predictive power for weakly augmentation, unlabeled images to generate pseudo-labels and filter them using a threshold. These pseudo-labels are then used to supervise the training of strongly augmented prediction results. Experimental results demonstrate the effectiveness of the method. In contrast, Yang et al. [27] argued that in the preparation phase, despite manually established confidence thresholds, erroneous pseudo-labels remain prevalent and have the potential to hinder the learning process of student models. Therefore, they used an improved data augmentation technique to obtain knowledge from unlabeled images while introducing additional information to enhance the learning process.

In addition, there are studies to rectify pseudo-labels more flexibly through auxiliary networks than filtering pseudo-labels through thresholds. In [28] by Mendel et al. the network is evaluated by an auxiliary network in addition to receiving supervised training with annotated data. The auxiliary correction network is trained using annotated data to determine accurate predictions and correct inaccurate ones. However, training such networks effectively under Semi-Supervised Learning conditions is a considerable challenge. Auxiliary networks tend to quickly overfit labeled data, resulting in the undergeneralization of the correction network. Kwon and Kwak [29] proposed ELN avoid this problem by multiple auxiliary decoders. However, implementing the ELN method leads to increased training overhead due to numerous decoders.

Although the auxiliary networks have some drawbacks, the above study confirms the potential of the auxiliary model in correcting pseudo-labels.

C. Federated Semi-Supervised Learning

Federated Semi-Supervised Learning combines the advantages of Federated and Semi-Supervised Learning to protect site data privacy while reducing the cost of annotation, which is of particular importance in healthcare applications.

Two main types of Semi-Supervised Learning configurations exist in the Federated environment.

One type considers that some sites are fully labeled while others contain unlabeled images, e.g. [30], [31], and [32]. In this scenario, Liang et al. [30] proposed to refine several sub-consensus models by random sampling of sites, and then aggregating these sub-consensus models into the global model for collaborative training. However, Li et al. [31] argued that the sub-consensus model is vulnerable to the imbalance of the category distribution. Therefore, adaptive threshold classes that maintain the information transfer of unlabeled data and design class balance are proposed to study Federated Semi-Supervised Learning. Fan et al. [32] efficiently train the model by creating a uniform data space from sparse and skewed labeled data distributions across all participating sites.

However, this assumption does not hold in practical scenarios. On one hand, users often lack a sufficient supply of labeled data [33] as labeling data requires both time and domain knowledge. On the other hand, servers typically hosted by organizations are more likely to acquire labeled data than individual users. For instance, in cross-institutional scenarios where multiple medical research institutes collaborate on disease diagnosis, a disease may be newly discovered by one institute, leaving other institutes without labeled samples [33]. In such situations, Zhang et al. [34] observed that the use of consistency regularization loss and group regularization in Semi-Supervised Learning can reduce gradient diversity and improve test accuracy. Diao et al. [35] proposed alternative training methods of fine-tuning the global model with labeled data and generating pseudo-labels with the global model to improve model performance. Jiang et al. [36] solved the class distribution imbalance problem among unlabeled clients by using class scaling information to improve the training of clients. Jeong et al. [37] used new inter-client consistency loss and parameter decomposition to improve the combination of Federated and Semi-Supervised Learning. Zhang et al. [38] proposed a model parameter mixing and dynamic weight adjustment strategy to improve the efficiency of UAV image recognition. Pei et al. [39] explored the potential association

Algorithm 1 Model-Heterogeneous Semi-Supervised Federated Learning

Input: Public dataset D_p; Local datasets $\{D_i\}_{i=1}^{K}$; Communication Rounds T;
Output: Local models $\{\theta_i\}_{i=1}^{K}$; Server Model θ_p

for *Round* $t = 1 \dots T$ **do**
  `/* Regularity Condensation at the server */`
  $\theta_p^t \longleftarrow$ **Re-Condense**$(\theta_p^{t-1}, \{\theta_i^{t-1}\}_{i=1}^{K}, D_p)$;
  for *Site* $k = 1 \dots K$ *Parallelly* **do**
    `/* Regularity Fusion to the site */`
    $\theta_k^t \longleftarrow \theta_k^{t-1} - \alpha \nabla_\theta L_{kl}(D_p, \theta_k^{t-1}, \theta_p^t)$;
    `/* Local Semi-Supervised Learning */`
    $\theta_k^t \longleftarrow$ **Local-SSL**(θ_k^t, D_k, D_p);
  end
end
return;
Function `Re-Condense(`$\theta_p^{t-1}, \{\theta_i^{t-1}\}_{i=1}^{K}, D_p$`)`**:**
  `/* Personalized selection regulations based on Grade */`
  $\tilde{\theta}_s^{t-1} = \{Max[Grade(x_i^p,\theta_1^{t-1}),...Grade(x_i^p,\theta_k^{t-1})]\}_{i=1}^{N_p}$;
  `/* Use KL loss to condense regulations */`
  $\theta_P^t \leftarrow \theta_P^{t-1} - \alpha \nabla_\theta L_{kl}(D_p, \theta_P^{t-1}, \tilde{\theta}_s^{t-1})$;
  return;
;
Function `Local-SSL(`θ_k^t, D_k, D_p`)`**:**
  for *Site* $k = 1 \dots K$ *Parallelly* **do**
    `/* Personalized selection of global regulations to fusion */`
    $\theta_k^t \leftarrow \theta_k^t - \alpha \nabla_\theta [L_{sup}(D_p,\theta_k^t) + L_{unsup}(D_k,\theta_k^t)]$;
  end
  return;
;

between labeled and unlabeled records through a knowledge transfer mechanism. And a specially designed subgraph aggregation capsule network is used to capture various malicious behaviors.

The above studies organically combine Federated Learning with Semi-Supervision. However, most of these studies tend to solve classification tasks and are often set in homogeneous scenarios without taking into account the individual needs of the site, especially the structural specificity issues. In this paper, Federated Semi-Supervised Learning is applied to Medical Image Segmentation, using autonomous knowledge transfer to protect privacy and improve model heterogeneity. The **SA** module and **RPG** module are used simultaneously to improve the efficiency of Local Semi-Supervised training, achieving a perfect balance of overall performance and annotation costs.

III. Method

We formulate the problem in the first section and explain our solutions next. The general process is shown in Alg. 1.

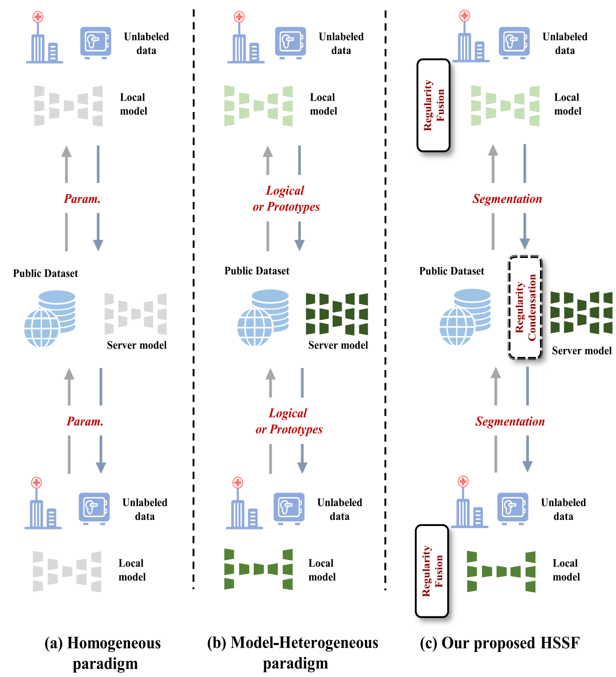

Fig. 1. Overview of Federated Learning paradigms. **(a)** Homogeneous paradigm, which requires that the site and server have the same model structure and are updated by parameter averaging. **(b)** Model-Heterogeneous paradigm, which uses public datasets as a medium and delivers knowledge by aligning client and server predictions. **(c)** Our proposed **HSSF**. Unlike the previous heterogeneous paradigm, we propose *Regularity Condensation* and *Regularity Fusion* to allow models to make autonomous judgments and adaptively select the optimal knowledge to learn, improving learning quality and efficiency.

A. Problem Formulation

The image segmentation task for Model-Heterogeneous Semi-Supervised Federated Learning is formulated as follows. Given K local sites and a server. The local dataset of the k-th site is denoted as $D_k = \{x_i^k\}_{i=1}^{N_k}$, with N_k samples. None of the local datasets are labeled and can only be used for local operations. Site k also maintains a local model with only locally visible structure and parameters. The model computation uses $f(\circ)$. $f(x_i, \theta_k)$ denotes the segmentation prediction result of x_i on the θ_k. A model θ_p is maintained on the server side. In addition, there is a public labeled dataset $D_p = \{x_i^p, y_i^p\}_{i=1}^{N_p}$, $|D_p| = N_p$, which is open to all sites. To minimize the communication overhead, D_p is usually much smaller than local datasets.

B. Heterogeneous Federated Learning

Our proposed Model-Heterogeneous Federated Learning requires two phases: *Regularity Condensation* and *Regularity Fusion*, as shown in Fig. 2. Rather than aggregating all local knowledge, the server first selectively condenses the regulations from the K local sites. Then, the server broadcasts these regulations to each site, and each site selectively condenses some of them according to its reality. This process uses D_p as a carrier of the regulations for transfer. The specific processes of *Regularity Condensation* and *Regularity Fusion* are described below.

1) Regularity Condensation: *Regularity Condensation* is the most critical process of HSSF, including two major steps,

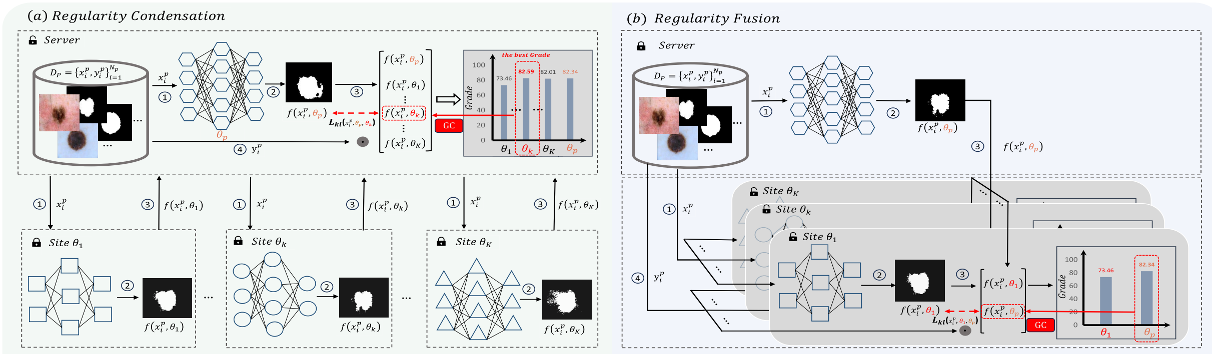

Fig. 2. Overview of Model-Heterogeneous Federated Learning. The framework consists of K sites and a server, each with a dataset D_k for **local use only** and **unlabeled**. A public dataset D_p is maintained on the server side. Where the lock indicates whether the parameter is trained. (a) ***Regularity Condensation***. In the training phase, the server is first updated by learning the optimal predictions $f(x_i^p, \theta_k)$ of all clients. (b) ***Regularity Fusion***. The updated server regenerates new predictions $f(x_i^p, \theta_p)$ and broadcasts them to all sites, which selectively learn and update the local model according to their reality. The forward process consists of four steps: ① the server and local sites obtaining publicly available data; ② the server and local sites use the current model to predict the segmentation result; ③ for ***Regularity Condensation***, aggregates all segmentation prediction results at the server; for ***Regularity Fusion***, each site receives the server's predicted segmentation result. ④ calculating the *Grade* and *GC* at the site or server.

determining the regulations that represent the capabilities of a model and saving the filtered regulations for learning.

We believe that regulations represent empirical knowledge distilled by models through data analysis, a concept inherently abstract and integral to the process of pattern recognition. For instance, a model can diagnose skin lesions by analyzing known or unknown factors such as asymmetry, irregular boundaries, color variations, and more. However, due to the black-box nature of deep learning, we cannot directly extract and employ these regulations.

Therefore, we propose deploying a global model on the server to indirectly teach these regulations by simulating the output for corresponding images. However, in a model-heterogeneous Federated Learning setting, each client possesses its own set of regulations for each image. To address this, we establish an evaluation criterion for indirectly filtering the optimal regulations. These criteria are subsequently integrated into the global model to facilitate knowledge transfer.

Based on this, we design the process shown in Fig. 2 (a), where site k first determines the logical segmentation predictions $\{f(x_i^p, \theta_k)\}_{i=1}^{N_p}$ on D_p and upload it to the server. At the same time, the server determines the current logical segmentation predictions $\{f(x_i^p, \theta_p)\}_{i=1}^{N_p}$. Due to differences in the learning ability of the models, not all the regulations are applicable to the current model. After determining the regulations of the model indirectly, we filter the regulations.

To this end, we calculate the $Grade$ by (1), let the server filter the regulations independently based on the $Grade$, compare it with its own regulations, and choose the optimal regulations for fusion.

$$\begin{aligned} Grade((x_i^p, y_i^p), \theta_k) &= mean(f(x_i^p, \theta_k) \odot y_i^p), \\ GC((x_i^p, y_i^p), \theta_k, \theta_p) &= Grade((x_i^p, y_i^p), \theta_k) \\ &> Grade((x_i^p, y_i^p), \theta_p). \end{aligned} \tag{1}$$

where $mean(\circ)$ serves to return the pixel-level average and $\odot$ implies the 'XNOR' operation. y_i^p denotes the segmentation mask corresponding to image x_i^p, provided in one-hot form.

Then we update the regulations with (2).

$$L_{kl}(D_p, \theta_p, \theta_k) = \sum_{i=1}^{N_p} f(x_i^p, \theta_k) \log[f(x_i^p, \theta_k)/f(x_i^p, \theta_p)] \bullet GC. \tag{2}$$

2) Regularity Fusion: After the $Regularity\ Condensation$ phase, we believe that the server has integrated the excellent regulations in the previous state. Next, as shown in Fig. 2 (b), the server needs to synchronize these regulations to the individual sites. Note that we use a personalized $Regularity\ Fusion$ approach here because of the differences in the local data distribution of the sites and the autonomy of the structural design.

Specifically, the server first computes $\{f(x_i^p, \theta_p)\}_{i=1}^{N_p}$ for the current state, and then the server broadcasts $\{f(x_i^p, \theta_p)\}_{i=1}^{N_p}$ to each site. Site k selects some of the best regulations autonomously by computing GC, calculating the distribution difference with $L_{kl}(D_p, \theta_k, \theta_p)$ and updating the parameter settings of θ_k.

We utilize Fig. 3 to visualize the selection process of $Regularity\ Condensation$ and $Regularity\ Fusion$. Through this visualization, we can intuitively observe how the server and the sites achieve improved personalization by actively filtering and condensing certain knowledge during aggregation.

C. Local Semi-Supervised Learning

Model-Heterogeneous Federated Learning solves the problem of knowledge transfer, and next, we will explore how to effectively use local unlabeled data. We provide a new training paradigm to solve this problem, as shown in Fig. 4. We first perform local supervised training on D_p, and name it Labelled Local Training. The main purpose of this process is to obtain a mature and usable **SA** module, and then perform Unlabelled Self-training on local data. The **SA** confidence is used to correct the pseudo-labels and improve the local applicability of the model. Details are given below.

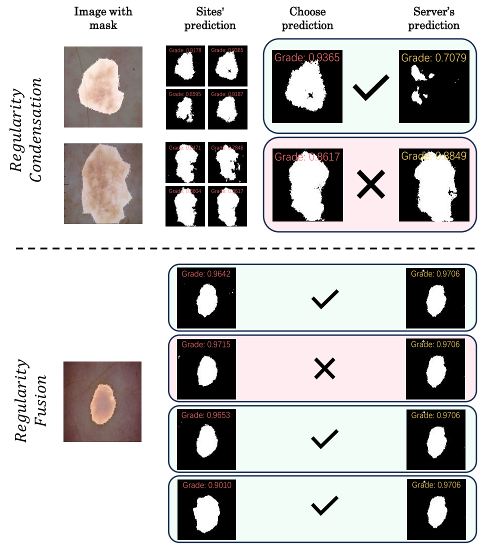

Fig. 3. Visual comparison of *Regularity Condensation* and *Regularity Fusion*. In this scenario with $K = 4$, **Image with mask** denotes an example where the mask covers the sample image. **Sites' prediction** refers to the predictions for all sites in the current image, along with their corresponding *Grade* (located at the center above the image). **Choose prediction** means selecting the best prediction based on the *Grade*. Similarly, **Server's prediction** includes the server's predictions and their corresponding *Grade*. In the *Regularity Condensation* part, the server evaluates the *Grade* to decide whether to simplify the regulations associated with the current image. This choice depends on the scores received. In contrast, in the *Regularity Fusion* part, each site decides whether to combine the server's regulations. This decision involves evaluating the server's regulations and deciding whether to integrate them.

1) Labelled Local Training: The local segmentation model here primarily corresponds to the structure of DeepLabv3+ [40]. We design the attention-based **SA** module based on this structure, and the details are shown in Fig. 4 (a). This module makes a self-assessment of the prediction for each position by considering the intermediate activation of the model and the prediction logic together to use local unlabeled data effectively certainly. The training process is as follows:

For the labeled data $(x_i^p, y_i^p) \in D_p$, We mainly supervise the training of the model from two aspects: segmentation prediction p_i^p and **SA** confidence a_i^p. Like other supervised training paradigms, we use a combined loss based on IoU and Cross-Entropy, as shown in (3), using y_i^p to supervise p_i^p to update the model.

$$\begin{aligned} L_{iou}(p_i^p, y_i^p) &= 1 - [p_i^p \cdot y_i^p]/[p_i^p + y_i^p - p_i^p \cdot y_i^p], \\ L_{bce}(p_i^p, y_i^p) &= y_i^p \log(p_i^p) + (1 - y_i^p)\log(1 - p_i^p), \\ L_{seg}(p_i^p, y_i^p) &= L_{iou} + L_{bce}. \end{aligned} \tag{3}$$

For the **SA** confidence a_i^p, as shown in (4), where we calculate the $confide(p_i^p, y_i^p)$, and use $confide(p_i^p, y_i^p)$ to supervise a_i^p to calculate the cross-entropy loss.

$$\begin{aligned} confide(p_i^p, y_i^p) &= p_i^p \odot y_i^p, \\ L_{confide}(a_i^p, p_i^p, y_i^p) &= L_{bce}(a_i^p, confide(p_i^p, y_i^p)). \end{aligned} \tag{4}$$

The loss of the supervised training process is calculated by (5), where μ is a hyperparameter to control the weight of the two.

$$L_{supervised} = \mu L_{confide} + L_{seg}. \tag{5}$$

2) Unlabelled Self-Training: In this section, we will augment the model with local unlabeled data. We explore this in two ways: consistency regularization, taking inspiration from [41], and we add feature perturbations to the most fundamentally strong and weak consistency; Self-training, we propose the **RPG** module to improve the correctness of pseudo-label with **SA** confidence as a clue. The specific process is as follows:

As shown in Fig. 4 (b), for private data $x_i^k \in D_k$, we first apply weak and strong perturbations to x_i^k, generating $\dot{x}_i^k$ and $\ddot{x}_i^k$, respectively. We also apply feature perturbations on the features that were extracted from $\dot{x}_i^k$, $\dot{p}_i^k$, $\ddot{p}_i^k$, $\hat{x}_i^k$ denote the segmentation prediction corresponding to different perturbations, respectively, and $\dot{a}_i^k$, $\ddot{a}_i^k$ denote the current model's **SA** confidence for $\dot{p}_i^k$, $\ddot{p}_i^k$.

We recommend using the **RPG** module to improve the accuracy of the pseudo-label. According to the initial intention of the **SA** module, when the confidence level of a specific position is too low, we believe that the prediction result using the opposite value will be more convincing than the current one, and when the confidence level of the sample as a whole is too low, we have a reason to discard the sample. The precise procedure is this: first, we compare $\dot{a}_i^k$, $\ddot{a}_i^k$, and then filter $\dot{p}_i^k$, $\ddot{p}_i^k$ using the comparison results as an index to create a confidence γ_i^k and a pseudo-label pl_i^k. Then γ_i^k is compared with a predetermined threshold ε, and if the part is less than the ε, it will be inverted to produce the final pseudo-label pl_i^k; if the overall mean is less than the threshold ζ, it will be discarded.

The training function (6) illustrates the two portions of the training function, one of which is based on consistency and the other on the created pseudo-label. We initially assess the confidence level γ_i^k for image x_i^k. By comparing the difference between the results of the strong perturbation $\ddot{p}_i^k$ and the weak perturbation $\dot{p}_i^k$, respectively, using the created pseudo-label pl_i^k, we supervised the training of the entire model. Additionally, by calculating the difference between $\hat{p}_i^k$ and $\dot{p}_i^k$, we improve the consistency of the model.

$$\begin{aligned} L_{mse}(\hat{p}_i^k, \dot{p}_i^k) &= (\hat{p}_i^k - \dot{p}_i^k)^2, \\ L_{pseg}(\dot{p}_i^k, \ddot{p}_i^k, pl_i^k, \gamma_i^k) &= [L_{bce}(\dot{p}_i^k, pl_i^k) + L_{bce}(\ddot{p}_i^k, pl_i^k)] * (\gamma_i^k > \zeta), \\ L_{unsupervised} &= L_{mse} + L_{pseg}. \end{aligned} \tag{6}$$

IV. Experimental

A. Datasets

The initial scenario focuses on the problem of the Polyp Lesion Segmentation. The selected datasets adhere to a non-independently distributed (Non-IID) design and comprise five publicly available datasets with varying distributions: Kvasir [42], CVC-ClinicDB [43], CVC-ColonDB [44], CVC-300 [45], and EndoTectETIS [46]. Specifically, the ColonDB dataset [44] simulates a public dataset (D_p) and includes a total of 380 images. The other four datasets are simulated to represent private datasets for individual clients, each containing a varying number of images ranging

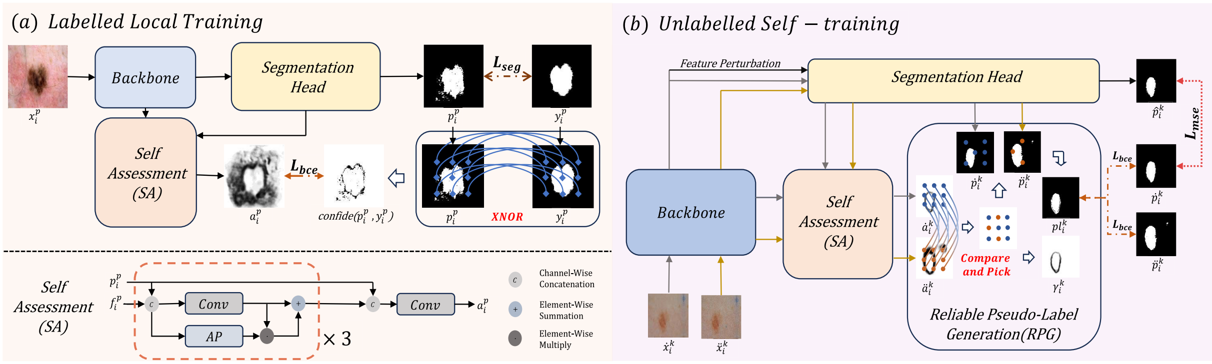

Fig. 4. Overview of Local Semi-Supervised Learning (**LSSL**), including two cascaded steps that are (a) ***Labelled Local Training***, where the local model is trained by supervised learning of D_p and supervised training of the **SA** module using the predicted and labeled *XNOR* results. f_i^p represents features extracted by backbone, *conv* represents convolution operations, and *AP* represents average pooling. (b) ***Unlabelled Self-training***, where the **RPG** module generates a reliable pseudo-label pl_i^k based on the prediction confidence scores $\dot{a}_i^k$, $\ddot{a}_i^k$ of the SA module. Then, the local model training is supervised using pl_i^k corresponding to the local unlabeled data x_i^k.

from 300 to 612. The private datasets undergo a partitioning process, during which they are allocated into separate subsets, including training, validation, and test sets. The allocation ratio is set at 0.8:0.1:0.1.

The subsequent scenario revolves around simulating the treatment of skin lesions, following an IID setup, with the primary utilization of the ISIC-2018 dataset. The International Skin Imaging Collaboration (ISIC) curated and administered this data corpus in 2018, consisting of 2,694 image files and corresponding labels. The images exhibit varying resolutions, ranging from 720 × 540 to 6708 × 4439 pixels. A cohort of 100 individuals was randomly chosen and designated as D_p, with their corresponding annotations being utilized for analysis. The remaining individuals were assigned to four discrete locations according to the divisions of 200, 400, 800, and 1194 as D_1, D_2, D_3, and D_4, and were considered as private data for each corresponding location.

We employed t-SNE to visualize the grayscale feature distributions of various datasets under homogeneous and heterogeneous settings separately. The results are illustrated in Fig. 5.

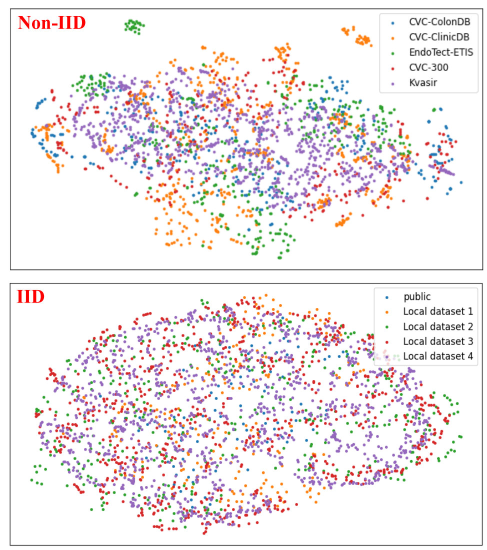

Fig. 5. Dataset Feature Distribution Overview. We separately recorded the grayscale feature distributions of different datasets under Non-IID (upper) and IID (lower) settings. Additionally, we employed t-SNE for feature visualization.

B. Evaluation Metrics

In order to assess the efficacy of the proposed model in terms of segmentation performance, we employ two widely-used metrics, namely the Dice coefficient($Dice$) and the 95^{th} percentile of the Hausdorff distance of the boundaries ($HD95$). The present study employs metrics to evaluate the effectiveness of the segmentation model quantitatively. Precisely, segmentation performance is gauged by measuring area-based metrics, such as the Dice coefficient, which are expected to be higher in instances of superior performance. Conversely, boundary-based metrics, such as Hausdorff distance at 95% percentile, are anticipated to exhibit lower values in effective segmentation models. The procedure for calculating these two evaluation metrics is described in 7. where p_i denotes the segmentation prediction, y_i denotes the corresponding expert annotation and $\hat{\max}\{\cdot\}$ refers to the calculation of the 95^{th} percentile of the distances.

In addition, we conduct a paired samples t-test and report the results in the specified locations in Table I and Table II. We compare the statistical significance of the **Dice** performance between ours and other methods in both homogeneous and heterogeneous settings. A p-value less than 0.05 was considered indicative of a significant improvement in performance.

$$\begin{aligned} Dice(p_i, y_i) &= 2 * \frac{|p_i * y_i|}{|p_i| + |y_i|}, \\ h(p_i, y_i) &= \hat{\max_{a \in p_i}}\{\min_{b \in y_i} ||a - b||\}, \\ HD95(p_i, y_i) &= \max\{\mathrm{h}(p_i, y_i), \mathrm{h}(y_i, p_i)\}. \end{aligned} \tag{7}$$

C. Implementation Details

In order to compare our results with those of other works fairly, we simulate four sites in a heterogeneous setting. Site

TABLE I

COMPARING FEDERATED SEMI-SUPERVISED LEARNING METHODS FOR POLYP LESION SEGMENTATION. WE EVALUATE DIFFERENT METHODS FOR POLYP LESION SEGMENTATION IN FEDERATED SEMI-SUPERVISED LEARNING. WE PROVIDE MEAN AND STANDARD DEVIATION VALUES FOR *Dice* AND *HD*95 SCORES ON LOCAL DATASETS, ALONG WITH THE CORRESPONDING P-VALUE FROM THE SIGNIFICANCE TEST

	Hetero.	p-value ($\times 10^{-5}$) ↓	**Dice** (%) ↑					**HD95** $(pix.)$ ↓				
			c_1	c_2	c_3	c_4	**Average**	c_1	c_2	c_3	c_4	**Average**
Mean Teacher [49]	✗	0.027	17.23 ± 11.89	17.44 ± 21.70	11.04 ± 7.45	27.88 ± 18.89	21.06 ± 17.78	250.4 ± 41.26	249.7 ± 75.93	275.5 ± 50.91	207.1 ± 53.04	234.9 ± 60.03
FixMatch [26]	✗	0.779	36.05 ± 39.65	32.00 ± 36.59	66.66 ± 38.62	41.35 ± 38.17	41.60 ± 39.48	66.37 ± 72.49	76.22 ± 65.45	**17.67 ± 23.30**	83.90 ± 71.18	68.51 ± 68.89
UniMatch [41]	✗	11.95	35.37 ± 39.21	34.34 ± 37.14	71.27 ± 34.57	52.32 ± 37.01	47.14 ± 39.10	83.77 ± 73.15	113.2 ± 94.99	26.71 ± 51.33	76.22 ± 65.30	76.21 ± 74.04
ELN [29]	✗	2.207	40.37 ± 31.92	27.45 ± 33.24	37.56 ± 35.44	53.10 ± 27.96	43.27 ± 32.25	92.66 ± 89.60	106.5 ± 77.36	66.95 ± 76.55	90.14 ± 65.19	90.70 ± 74.86
SemiFL [35]	✗	2408	54.59 ± 30.52	42.56 ± 35.15	73.86 ± 29.16	65.21 ± 27.14	59.59 ± 31.21	97.81 ± 68.01	119.6 ± 78.71	65.17 ± 81.08	84.00 ± 61.86	91.30 ± 70.57
LSSL (ours)	✗	-	**56.79 ± 34.16**	**56.07 ± 37.15**	**74.16 ± 32.38**	**65.88 ± 27.12**	**62.83 ± 32.04**	**66.00 ± 56.79**	**60.73 ± 71.01**	27.36 ± 27.16	**63.94 ± 54.47**	**59.20 ± 56.37**
FedMD [9]	✓	177.1	52.47 ± 28.71	39.81 ± 35.82	71.58 ± 25.19	36.34 ± 27.26	45.81 ± 31.23	90.38 ± 63.44	**70.61 ± 60.93**	37.13 ± 32.99	110.6 ± 59.83	89.68 ± 62.96
KT-pFL [11]	✓	2.621	58.37 ± 30.71	32.22 ± 30.05	72.38 ± 27.44	52.60 ± 32.68	53.19 ± 32.94	70.88 ± 62.15	117.3 ± 75.58	34.65 ± 51.75	84.12 ± 66.98	79.11 ± 68.81
FedDistill [50]	✓	1.284	28.12 ± 25.82	30.65 ± 27.99	47.77 ± 36.86	55.38 ± 24.86	42.85 ± 29.98	118.4 ± 71.86	133.6 ± 71.26	70.49 ± 77.00	104.7 ± 60.08	109.2 ± 68.93
FedProto [13]	✓	0.005	39.44 ± 26.19	22.47 ± 27.65	47.36 ± 29.05	51.19 ± 22.52	42.60 ± 27.23	108.6 ± 70.82	110.5 ± 52.71	41.85 ± 31.66	127.2 ± 55.35	109.8 ± 63.09
RHFL [12]	✓	0.819	56.83 ± 26.56	40.03 ± 33.05	68.71 ± 28.81	63.12 ± 27.91	58.18 ± 29.85	68.36 ± 59.03	86.26 ± 69.30	37.77 ± 58.60	69.47 ± 53.56	68.00 ± 59.57
HSSF (ours)	✓	-	**66.35 ± 27.78**	**42.73 ± 37.20**	**78.62 ± 28.41**	**72.82 ± 24.58**	**66.65 ± 30.53**	**51.07 ± 50.20**	83.24 ± 80.70	**25.30 ± 49.20**	**54.15 ± 52.68**	**54.01 ± 58.52**

TABLE II

COMPARING FEDERATED SEMI-SUPERVISED LEARNING METHODS FOR SKIN LESION SEGMENTATION. WE EVALUATE DIFFERENT METHODS FOR SKIN LESION SEGMENTATION IN FEDERATED SEMI-SUPERVISED LEARNING. WE PROVIDE MEAN AND STANDARD DEVIATION VALUES FOR *Dice* AND *HD*95 SCORES ON LOCAL DATASETS, ALONG WITH THE CORRESPONDING P-VALUE FROM THE SIGNIFICANCE TEST

	Hetero.	p-value ($\times 10^{-5}$) ↓	**Dice** (%) ↑					**HD95** $(pix.)$ ↓				
			c_1	c_2	c_3	c_4	**Average**	c_1	c_2	c_3	c_4	**Average**
Mean Teacher [49]	✗	0.724	69.39 ± 21.95	69.76 ± 20.83	73.59 ± 21.43	77.29 ± 19.44	74.39 ± 20.59	66.39 ± 59.05	54.49 ± 47.08	48.92 ± 46.58	48.01 ± 48.25	50.70 ± 48.44
FixMatch [26]	✗	51.46	68.86 ± 36.09	76.04 ± 27.60	74.70 ± 31.22	79.10 ± 24.84	76.49 ± 28.28	46.57 ± 64.48	**35.13 ± 40.95**	38.87 ± 50.46	39.80 ± 49.90	39.31 ± 49.80
UniMatch [41]	✗	495.6	76.81 ± 27.48	78.54 ± 26.34	80.26 ± 24.65	81.11 ± 21.51	80.12 ± 23.65	51.30 ± 67.48	41.03 ± 54.42	34.80 ± 47.52	37.15 ± 45.08	38.13 ± 49.22
ELN [29]	✗	0.223	73.05 ± 28.29	76.16 ± 22.03	72.10 ± 28.90	75.90 ± 23.00	74.55 ± 25.16	38.35 ± 37.71	39.85 ± 42.00	48.21 ± 52.74	43.24 ± 45.46	43.88 ± 46.69
SemiFL [35]	✗	171.9	77.13 ± 18.95	78.09 ± 21.05	79.52 ± 19.14	82.71 ± 15.51	80.59 ± 17.88	50.48 ± 50.47	46.52 ± 47.40	45.20 ± 49.55	37.87 ± 39.43	42.42 ± 44.81
LSSL (ours)	✗	-	**86.59 ± 7.43**	**81.06 ± 19.20**	**81.47 ± 19.19**	**84.94 ± 15.23**	**83.40 ± 16.80**	**19.35 ± 11.91**	38.28 ± 42.51	**29.81 ± 37.31**	**27.43 ± 30.11**	**29.21 ± 33.85**
FedMD [9]	✓	1.414	70.41 ± 19.43	70.94 ± 21.04	73.17 ± 21.23	77.86 ± 19.45	74.78 ± 20.36	49.43 ± 44.35	47.22 ± 40.98	52.20 ± 48.42	42.75 ± 39.77	46.84 ± 43.06
KT-pFL [11]	✓	0.039	58.18 ± 24.06	51.52 ± 22.14	50.91 ± 32.13	65.35 ± 25.59	58.23 ± 27.91	68.12 ± 38.03	80.81 ± 26.56	138.6 ± 74.30	62.86 ± 47.50	89.24 ± 63.58
FedDistill [50]	✓	10.61	66.05 ± 29.38	75.62 ± 21.63	75.74 ± 23.49	75.75 ± 22.84	74.98 ± 23.42	63.02 ± 56.52	43.79 ± 46.09	45.41 ± 50.27	49.34 ± 46.06	48.34 ± 48.19
FedProto [13]	✓	21.10	68.01 ± 30.80	76.95 ± 20.20	69.40 ± 23.88	75.92 ± 21.03	73.46 ± 22.81	56.04 ± 51.45	50.14 ± 49.96	58.60 ± 51.81	60.95 ± 56.54	58.19 ± 53.60
RHFL [12]	✓	0.819	65.89 ± 34.11	67.92 ± 22.21	73.43 ± 19.21	77.12 ± 21.45	73.70 ± 22.33	**44.57 ± 46.75**	48.69 ± 39.15	46.92 ± 39.38	38.53 ± 33.77	43.12 ± 37.39
HSSF (ours)	✓	-	**72.18 ± 27.84**	**77.91 ± 19.79**	**77.34 ± 22.60**	**81.83 ± 16.16**	**79.11 ± 20.00**	51.10 ± 45.53	**41.74 ± 38.20**	**32.30 ± 37.91**	**30.35 ± 34.80**	**34.18 ± 37.39**

c_1 and site c_2 use ResNet18 [47] as the backbone, while the other two sites, site c_3 and site c_4, use ResNet34 as the backbone. We use ResNet101 on the server side to extract the intermediate activation. In the case of homogeneous models, it is worth noting that all sites utilize the ResNet34 architecture as the backbone. The transmission of the models is realized by obtaining the mean value of the model parameters.

Our study employs a standardized image size of 384×384, and an identical image perturbation approach [26]. Further, we utilize a feature perturbation method in Unimatch [41] that involved a 50% probability of channel discard using Dropout2d(0.5) in PyTorch [48]. The AdamW optimizer is the default option, with an initial learning rate of 0.0001. We establish the unmarked threshold ε at 0.3 and ascertain the confidence threshold ζ to be 0.75. The parameter of confidence, denoted as μ, is set to 4.

D. Comparison With SOTA Homogeneous Federated Learning Methods

In this section, we conduct comparisons under homogeneous conditions with the following specific setup: we select some widely recognized Semi-Supervised methods, namely FixMatch [26], UniMatch [41], ELN [29], and our proposed LSSL. These methods were combined with FedAvg [7] in a straightforward manner, where site-side local training utilized both D_p and its private dataset. After training, local parameters were uploaded, and the server performed parameter-averaging updates, which were then distributed back to the clients for further fine-tuning. Additionally, we also chose the SemiFL framework [35], specifically designed for addressing Federated Semi-Supervised Learning under homogeneous conditions, for comparison purposes.

The experimental results are presented in the upper part of Table I and the upper part of Table II, respectively. We conduct statistical significance tests comparing other methods to LSSL, and the results indicate that our approach exhibits significant advantages in homogeneous scenarios. Additionally, we have provided a visual comparison in Fig. 6.

The current results indicate that the effectiveness of model homogenous training methods has a significant impact when the public dataset and local dataset exhibit non-IID characteristics. The data in Table I corroborates this observation. However, when the public and local datasets follow an IID distribution, employing model homogenous training techniques such as parameter averaging can effectively enhance collaborative training outcomes. The outcomes are depicted in Table II.

Despite this fact, our approach maintains superiority over other SOTA methods.

E. Comparison With SOTA Heterogeneous Federated Learning Methods

In this section, under the premise of model heterogeneity, we compare HSSF with several SOTA model-heterogeneous Federated Learning paradigms, including FedMD [9],

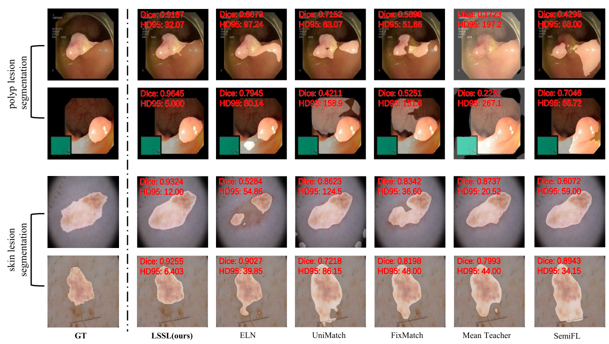

Fig. 6. Visual comparisons were performed in a number of representative Semi-Supervised Learning algorithms incorporating Homogeneous Federation Learning and Federated Semi-Supervised Learning algorithms. We selected skin lesion segmentation images and polyp segmentation images for different scenes, respectively, and we utilize the color red in the upper left corner of each visualization to highlight the respective Dice and HD95 scores.

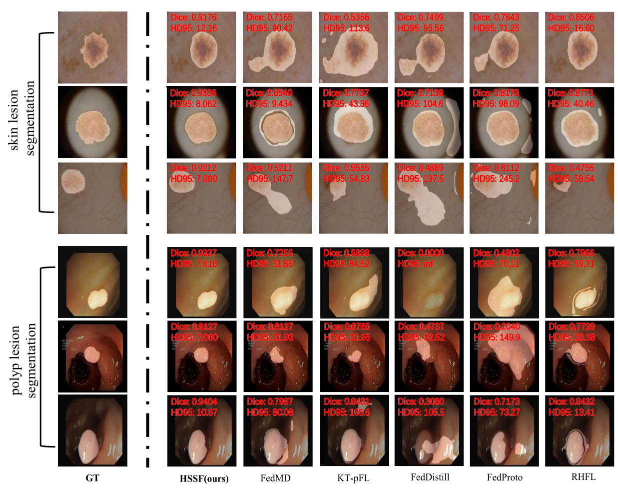

Fig. 7. Visual comparison is performed in some representative Model-Heterogeneous Federated Learning algorithms. We selected representative and challenging skin lesion segmentation images and polyp segmentation images, respectively, and we utilize the color red in the upper left corner of each visualization to highlight the respective Dice and HD95 scores.

KT-pFL [11], FedDistill [50], FedProto [13], and RHFL [12]. Since these methods are already examined within the setting of a classification assignment, we replace the classification demonstration with a segmentation demonstration to create a reasonable comparison. The specific settings are as follows: in order to accurately assess the performance of these methods with respect to model heterogeneity, we will uniformly employ a local update strategy, namely LSSL. Additionally, we will ensure that other factors that could potentially impact the experimental results, such as data partitioning, remain consistent.

We examine our model on the Polyp Lesion Segmentation and Skin Lesion Segmentation datasets, and the test results are displayed in the lower part of Table I and the lower part of Table II, respectively. We have provided a visual comparison in Fig. 7. In addition, the significance tests have confirmed the effectiveness of HSSF. Furthermore, experimental results demonstrate that when there are distributional differences between public and local datasets, Table I shows that the information provided by the direct accumulation of predictions or prototypes does not provide enough information to build a complete system. This also confirms the effectiveness of our autonomous selective knowledge transfer process, and our approach yields SOTA findings for boundary indicators in addition to regional evaluation indicators.

Furthermore, to further validate the adaptability of our method, we conduct comparisons under the distribution of IID between the public dataset and local datasets, as shown in Table II. Compared with the results in Table I, most of the methods are significantly improved, nevertheless, our method still achieves SOTA results.

F. Ablation Study

The present section analyzes the efficacy of individual modules within our architectural framework via the ablation learning technique. The aforementioned data analysis has been conducted utilizing the Dice coefficient, as illustrated in Fig. 8 (d). Concurrently, to directly depict the model's impact, we have provided visual representations of select representative images in Fig. 8 (e).

The baseline is established by computing the average results of supervised training for each site in the D_p. Subsequently, we supplement substantial indigenous unlabeled data while strictly limiting all communication. The resultant effect of incorporating the SA module alongside merging it with the RPG module was evaluated through rigorous testing. Dice has achieved an approximate 10% enhancement in polyp segmentation data and an approximate 5% enhancement in skin lesion segmentation. Additionally, this finding serves as evidence for the effectiveness and suitability of our proposed Semi-Supervised method in tackling both types of problems. Subsequently, we implemented a knowledge transfer mechanism into the Semi-Supervised method, employing a heterogeneous training strategy. The present model demonstrates a notable enhancement in performance, with up to a 3% improvement compared to the Semi-Supervised approach when applied to the polyp segmentation dataset.

We investigate the influence of various unmarked thresholds (denoted as ε), confidence parameter (denoted as μ), and the confidence thresholds (denoted as ζ) on HSSF, and the results are depicted in Fig. 8 (a), (b), (c).

Our findings revealed that ε has a significant impact on the segmentation performance. Setting ε too high leads to the filtration of a substantial amount of useful information, which is detrimental to model training. Conversely, when ε is set too low, the model can be misled by false labels, hindering its improvement. Experimental results demonstrate that the optimal performance is achieved when the threshold ε is set to 0.3.

The role of μ is to supervise the model in maintaining consistent **SA** judgments on segmentation results. When μ is too low, the model may neglect the training of **SA**, thereby affecting the utilization of unsupervised data. Conversely, when μ is excessively high, the model tends to be overly cautious, further impacting segmentation performance. The function of ζ is akin to filtering unlabeled data. A high ζ value results in a significant loss of knowledge, while a low ζ value exposes the model to negative knowledge, influencing

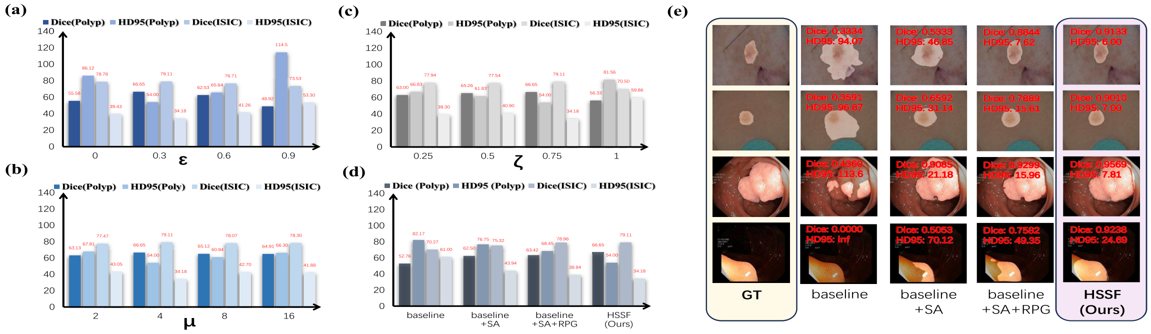

Fig. 8. Impact of varied thresholds and ablation experiments on Skin Lesion and Polyp Segmentation using the **HSSF** framework. (a) Effect of different ε values on segmentation using the **HSSF** framework. (b) Impact of varied μ values on segmentation using the **HSSF** framework. (c) Influence of diverse ζ values on segmentation using the **HSSF** framework. (d) Evaluation of modules within the **HSSF** framework on segmentation experiments. (e) Visual comparison of the **HSSF** framework's modules on Skin Lesion and Polyp Segmentation. We have utilized the Dice and HD95 metrics for comparative analysis. To highlight Dice and HD95 scores, we have placed them in red at the upper left corner of each visualization.

TABLE III

Comparison of Resource Utilization in Various Federated Learning Scenarios. We Give the Memory Consumption(Mem. Con.), Local Update Time(L.U. Time), Transmission Time(Trans. Time), Global Update Time(G. U. Time), and Summary Time(Sum. Time) Required for a Single Epoch Update

	Skin Lesion Segmentation					Polyp Lesion Segmentation				
	Mem. Con. (MB, ↓)	L. U. Time (s, ↓)	Trans. Time (s, ↓)	G. U. Time (s, ↓)	Sum. Time (s, ↓)	Mem. Con. (MB, ↓)	L. U. Time (s, ↓)	Trans. Time (s, ↓)	G. U. Time (s, ↓)	Sum. Time (s, ↓)
FedAvg(ResNet18)	81.71	137.55	16.34	**1.04**	**154.93**	81.71	97.20	16.34	**1.04**	**114.58**
FedAvg(ResNet34)	120.33	153.62	24.07	2.42	180.11	120.33	109.09	24.07	2.42	135.58
HSSF(ours)	**7.04**	**132.27**	**1.41**	24.45	158.13	**26.77**	**70.17**	**5.35**	55.15	130.67

its effectiveness. Through experimentation, it was found that the optimal model performance is achieved when μ is set to 4 and ζ is set to 0.75.

G. Resource Utilization Analysis

In this section, we conduct a fair comparison of computational and transmission costs to provide an intuitive demonstration of the practical feasibility of HSSF in real-world deployments, as shown in Table III. The comparison encompasses time and memory costs for both Skin Lesion Segmentation and Polyp Lesion Segmentation.

1) Memory Costs: Regarding memory costs, HSSF demonstrates a significant advantage, a consideration that has been integral to our design since its inception. In contrast to homogeneous approaches like FedAvg [7], which necessitate the transmission of the entire model's parameters, we employed a small-scale public dataset and transmitted only the model's prediction results to minimize transmission overhead. As illustrated in Table III, during Skin Lesion Segmentation, the transmitted file size is a mere 7.04 MB, accounting for only 0.09 of the ResNet18 parameter file size. For polyp segmentation, the transmitted file size remains quite small at 26.77 MB, significantly smaller than the ResNet18 parameter file size. This approach has been meticulously designed to accommodate bandwidth limitations, thereby extending the upper limits of scalable learning.

2) Time Costs: As for time costs, we have separately calculated the time required for local model updates during local training (L.U. Time), the time for communication and transmission between sites and the server (Trans. Time), the time for the server to aggregate client-transmitted data and perform global updates (G. U. Time), and the estimated total time for completing an entire training round (Sum. Time). It is worth noting that due to the additional *Regularity Condensation* and *Regularity Fusion* processes during training, HSSF indeed incurs more time during global updates. As indicated in Table III, HSSF consumes over ten times more time compared to similar methods. However, we believe this situation can be improved. Firstly, HSSF can synchronize the segmentation loss computation during *Regularity Fusion*, eliminating the need for training on labeled data during LSSL. This advantage is reflected in Table III, where our L.U. Time outperforms FedAvg [7]. Secondly, HSSF is more bandwidth-friendly, and we argue that, compared to computational power constraints, bandwidth limitations pose a more substantial bottleneck to the practical deployment of Federated Learning, especially in large-scale Federated Learning scenarios.

In summary, while HSSF introduces additional processes such as *Regularity Condensation* and *Regularity Fusion*, it significantly reduces memory consumption. In terms of time costs, given HSSF's integration of ResNet18 and ResNet34 structures, it offers approximately a 3.8% performance improvement compared to FedAvg(ResNet34), with a time reduction of approximately 3.6%, particularly in the polyp scenario.

V. Discussion

Medical Image Segmentation is a dependable fundamental for clinical diagnosis and pathology research by delimiting regions of interest in medical images and extracting pertinent features to support healthcare professionals in conducting precise diagnoses. Developing a proficient segmentation model requires a considerable volume of finely-detailed annotation data. Solely relying on the data available at a single medical institution is inadequate to fulfill the training requirements. Furthermore, due to ethical and privacy concerns, these institutions usually keep the data within their premises and impose restrictions on its accessibility.

The present study incorporates Heterogeneous Federated Learning in conjunction with Semi-Supervision. The approach of Heterogeneous Federated Learning enables collaborative training without compromising patient privacy while concurrently ensuring site personalization through aggregating the knowledge from diverse clients. Nonetheless, notwithstanding the availability of images, they remain deficient in supervised guidance. In light of this, we have adopted the Semi-Supervision principle to mitigate the labeling workload for medical practitioners. In contrast to prior research, our study exhibits superiority in two respects. Primarily, we conduct a comparative analysis of diverse approaches to knowledge transfer in the context of Heterogeneous Federated Learning and present our proposed methodology. The present study also details the development of a Semi-Supervised algorithm that has been streamlined for improved efficacy, drawing upon the extant literature to optimize its structure and performance. By implementing various experimental setups utilizing a skin lesion segmentation dataset and five polyp segmentation datasets, our findings indicate that our model exhibits robust segmentation performance with particular emphasis on the region segmentation metrics.

A majority of the prior literature concerning Federated Learning has been founded upon the utilization of parameter averaging. Parameter averaging necessitates that the client and server possess an identical model structure. The factor mentioned above exerts a significant constraint on the potential personalization of the site and impairs its overall efficacy. Hence, specific investigations have amalgamated knowledge distillation with Federated Learning, resulting in commendable performance outcomes of categorization endeavors. Taking inspiration from preceding research, we have incorporated their concepts in the context of segmenting data and introduced a novel approach to select mechanisms to achieve knowledge transfer autonomously. Our proposed method has been verified through experiments. In the present study, we uphold our preceding notion in the domain of Semi-Supervised Learning while putting forth a more straightforward approach to rectifying pseudo-label, particularly concerning the classical challenge of enhancing its dependability. The results of the comparison experiments indicate that our Semi-Supervised algorithm outperforms previous investigations to a significant extent.

Furthermore, in conjunction with Fig. 5, Table I, and Table II, we observe that when there is a significant disparity between the distribution of local and public datasets, the performance of local models is affected. For instance, in the polyp segmentation scenario, due to substantial distribution differences between CVC-ClinicDB [43] and EndoTectETIS [46] compared to the public dataset CVC-ColonDB [44], the effectiveness of c_1 and c_2 is compromised. In comparison, under the IID condition, the parameter averaging method proves advantageous, while for Non-IID, the knowledge distillation approach is more suitable. Second, due to the limited size of the annotated data, the segmentation bounds of the model prediction results exhibit suboptimal performance, which is conceivable. In subsequent research work, it may be necessary to establish a specific fusion mechanism in the knowledge distillation process to maintain the essential amount of information. Also, the model should pay more attention to the boundary information.

VI. Conclusion

In this study, we introduce a novel approach for Medical Image Segmentation, specifically aimed at leveraging a Model-Heterogeneous Semi-Supervised Federated Learning paradigm. Our proposed method is expected to have applicability beyond Medical Image Segmentation tasks.

Technically, we creatively introduced the **SA** module and the **RPG** module. By leveraging the XNOR-supervised training of the model with predictions and ground truth, we enhanced the model's self-evaluative capability, thereby improving the quality of pseudo-labels. To alleviate the burden of manual labeling, we conducted tests in a homogeneous scenario, successfully enabling training for all clients in the entire framework with only one annotated dataset. Additionally, to better meet practical needs, we innovatively proposed *Regularity Condensation* and *Regularity Fusion*. Building upon knowledge distillation, clients can not only autonomously construct their models but also selectively fuse regulations based on their specific preferences. This ensures personalization without compromising performance.

This study involved implementing a series of comparative experiments on a dataset consisting of skin lesions, as well as five additional datasets of polyps, aimed at evaluating and verifying the precision of their respective segmentation processes while also assessing their capacity for generalization.

Elaborate ablation analyses show that our HSSF architecture can attain high-quality segmentation outcomes by utilizing a limited quantity of annotated data while ensuring user confidentiality.

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